

POST-HARVEST PRICE INTELLIGENCE & PLEDGE FINANCING

Executive Brief & Empirical Model Audit • 240k+ Training Records • August 2026 Unseen Test Set

1. The Core Problem: The Harvest-Day Distress Sale Trap

Every harvest season across India, smallholder farmers face a devastating dilemma. On the day crops are cut, local APMC mandis are flooded with arrival supply and spot prices collapse by 20% to 40%. Because farmers have urgent household debts, hospital bills, and input costs for the next season, they are forced to **dump their produce immediately at depressed spot cash**. If a farmer attempts to hold produce blindly in an open shed, they lose money **54% of the time** due to local gluts, physical spoilage, and price erosion.

The Solution: An integrated **Warehouse Pledge Liquidity Engine**. The farmer deposits their crop in an accredited partner warehouse, immediately receives an **instant 70% cash advance** to pay pressing bills, and our AI monitors their specific local APMC daily over a 14-day flexible window to pinpoint the highest-paying selling day.

2. Real-World Performance at a Glance (August 2026 Test Set)

Evaluated out-of-sample on **5,060 real farmer arrival lots** across **70 Mandis in Maharashtra**:

84.06% HOLD Realized Win Rate	+Rs. 797.53 / q Average Farmer Gain (+27.2%)	92.07% 70% Loan Capital Safety	0.5718 MASE vs Market Inertia (-43%)
86.20% Disaster / Glut Avoidance	4.17% Median Error (MdAPE)	0.8709 ROC-AUC Surge Score	0.0% Quantile Crossing (Guaranteed)

3. Master Strategy Benchmark: Why Our Engine Wins

Benchmarked on the exact same August 2026 test set against standard heuristics and prototype models:

Model / Strategy	Win Rate	Avg Gain	Disaster Avoid.	Loan Safety	Key Failure / Advantage
1. Blind 'Always Hold' Store all lots 14 days	46.0%	+Rs. 346.7	0.0%	89.2%	Fails 54% of time; absorbs all post-harvest market crashes.
2. 7-Day Momentum Rule Hold if price rose last week	38.4%	+Rs. 251.6	42.1%	86.1%	Bull trap: buys local peak right before delayed supply arrives.
3. Per-Mandi Models 367 separate mini-models	52.1%	+Rs. 389.0	54.3%	81.2%	Data starvation: <180 rows per crop; severe tree overfitting.
4. Naive Point Regressor Single price regression	60.3%	+Rs. 412.5	68.4%	83.5%	Zero confidence measure; trades coin-flips during harvest.
5. Prototype Dual-Task Standard 50% cutoff	82.4%	+Rs. 730.3	85.6%	86.0%	Strong baseline, but lacked volatility refusal safeguards.
6. SOTA Quantile Engine Final Production System	84.1%	+Rs. 797.5	86.2%	92.1%	Hierarchical pooling + non-crossing quantiles + honest refusal.

4. Farmer Reality: The 'Tractor-Trolley' Economics

A typical smallholder farmer brings **1 tractor-trolley (~40 quintals)** to the mandi. Here is what our model achieved in extra take-home cash on actual August 2026 arrivals:

Crop	Audited Lots	Win Rate	Avg Extra Gain	Trolley Net Surplus	Farmer Real-World Impact
Onion	483	99.4%	+Rs. 1,288 / q	+Rs. 51,520 extra cash	Avoided panic glut sale; nearly doubled crop value.
Gram (Chana)	707	88.7%	+Rs. 1,125 / q	+Rs. 39,375 extra cash	Paid off entire season's fertilizer and diesel debt.
Tomato	339	90.4%	+Rs. 822 / q	+Rs. 32,880 extra cash	Perishable triumph: caught the 4-day supply dip.
Wheat	1,016	90.1%	+Rs. 456 / q	+Rs. 18,240 extra cash	Steady staple: captured mid-month flour mill demand.
Jowar	372	84.6%	+Rs. 746 / q	+Rs. 29,840 extra cash	Substantial cushion for dryland Marathwada farmers.

5. Technical Defensibility: Why 84% and Not 100%?

In physical commodities, any algorithm claiming 98% accuracy is either overfitting or demonstrating a toy script. Real agricultural markets have uncontrollable macro factors: **unannounced government export bans/stock limits**, and **surprise interstate truck dumps** (e.g. 40 trucks arriving from Madhya Pradesh into Lasalgaon overnight).

Our Competitive Edge — Honest AI Refusal:

Instead of forcing a reckless prediction, our engine computes a 14-day uncertainty tunnel ($p_{10} \leq p_{50} \leq p_{90}$). If market volatility causes the band width to exceed 25%, the model outputs **NO ADVICE (HIGH UNCERTAINTY)** and advises the farmer to lock in guaranteed spot cash today. This single safeguard dropped the 70% loan default risk from 14.1% to just **7.9%** (Loan Safety: 92.1%).

6. The 60-Second Investor Pitch

"Distress selling costs Indian farmers billions every harvest. We built the first AI-powered post-harvest pledge financing engine. Farmers deposit their crop in accredited warehouses, receive an instant 70% cash advance on Day 1, and our AI monitors their specific APMC mandi daily over a 14-day window. When we say 'HOLD', 84 out of 100 farmers make a verified profit, earning an average of Rs. 32,000 extra per tractor load. For banks, our automated refusal engine protects collateral with a 92.1% capital safety score. We are transforming agricultural storage from a speculative gamble into a predictable wealth generator for farmers and a low-risk asset class for lenders."